

VILSANDI, Estonia

WMO: 262140

Lat: 58.3828N

Lon: 21.8142E

Elev: 23

StdP: 14.68

Time Zone: 2.00 (EUE)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.9	13.5	2.5	6.3	9.4	8.3	8.4	15.3	36.4	33.3	33.2	33.1	9.1	80	0.697

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.2	77.8	68.1	74.7	66.7	71.8	65.1	69.7	75.4	67.9	73.3	66.0	70.8	9.7	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.5	101.1	73.4	65.6	94.7	71.4	63.8	88.8	69.4	33.7	75.8	32.2	73.4	30.7	71.0	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.3	28.4	25.4	DB	5.9	81.3	8.1	4.3	0.1	84.4	-4.6	86.9	-9.1	89.3	-15.0	92.4	(4)
(5)			WB	4.8	71.3	7.5	2.5	-0.6	73.1	-5.0	74.6	-9.2	76.0	-14.6	77.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.1	29.9	28.9	33.0	40.4	50.5	57.8	64.8	64.8	57.2	47.7	40.6	36.1		
	DBStd	13.85	8.15	8.02	5.56	4.96	6.52	5.01	4.80	4.21	4.62	5.92	5.77	7.07		
	HDD50	2935	622	591	526	292	69	2	0	0	7	113	281	432		
	HDD65	7016	1087	1011	991	737	452	225	62	53	234	536	731	897		
	CDD50	1509	0	0	0	5	85	236	458	459	224	42	0	0		
	CDD65	115	0	0	0	0	3	9	55	47	1	0	0	0		
	CDH74	351	0	0	0	0	12	40	181	118	0	0	0	0		
	CDH80	40	0	0	0	0	0	4	21	15	0	0	0	0		
	WSAvg	12.6	14.4	13.0	12.6	11.8	10.6	11.0	10.2	11.2	11.9	13.7	14.2	16.4		
	PrecAvg	22.9	1.8	1.3	1.3	1.1	1.3	2.0	2.0	2.9	2.3	2.5	2.5	2.0		
(15) Precipitation	PrecMax	31.3	4.4	2.8	2.7	2.3	2.5	3.8	4.1	5.8	4.6	4.3	4.5	4.5		
	PrecMin	15.7	0.4	0.1	0.4	0.2	0.3	0.2	0.1	0.1	0.4	0.4	0.2	0.7		
	PrecStd	3.9	0.9	0.6	0.6	0.6	0.6	0.8	1.1	1.6	1.1	1.2	1.1	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.9	40.5	49.2	61.9	75.5	79.2	82.8	81.7	70.1	60.0	51.7	48.1		
		MCWB	42.5	38.6	42.3	49.8	60.8	64.9	70.1	70.0	63.1	56.8	49.9	46.1		
	2%	DB	42.5	39.4	43.2	55.5	69.7	72.1	78.5	77.1	66.7	58.2	50.1	46.6		
		MCWB	41.1	37.9	39.4	46.7	59.6	63.7	69.0	68.2	61.0	55.4	48.3	44.8		
	5%	DB	40.9	38.4	40.5	51.0	64.8	67.6	75.5	73.9	64.8	56.8	48.9	45.2		
		MCWB	39.0	37.2	38.2	44.3	57.2	61.0	68.2	66.7	60.0	54.4	47.0	43.3		
	10%	DB	39.3	37.4	39.0	47.6	60.5	64.8	72.5	71.1	63.3	55.4	47.6	43.6		
		MCWB	37.8	36.5	37.2	43.0	54.6	59.7	66.2	65.0	59.1	53.0	45.7	41.7		
	0.4%	WB	43.0	39.4	43.2	50.3	63.9	67.6	72.7	71.7	64.7	57.9	50.3	46.9		
		MCDB	43.6	39.8	47.9	59.3	72.0	75.9	78.9	78.8	68.7	58.9	51.0	47.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.5	38.3	40.0	47.3	60.9	64.1	70.4	69.3	62.5	55.8	48.8	45.2		
		MCDB	42.3	38.9	42.5	53.9	67.3	70.9	76.3	74.6	65.5	57.4	49.8	46.4		
	5%	WB	39.4	37.4	38.6	45.1	57.9	62.1	68.7	67.6	61.0	54.7	47.4	43.6		
		MCDB	40.5	38.2	40.1	49.5	63.7	66.9	74.2	72.7	64.0	56.4	48.5	44.9		
	10%	WB	37.9	36.5	37.5	43.2	54.7	60.2	66.8	65.8	59.6	53.5	46.0	42.0		
		MCDB	39.1	37.3	38.7	46.7	60.3	64.0	71.9	70.2	62.6	55.3	47.3	43.4		
	5% DB	MDBR	5.7	6.1	6.3	7.9	9.4	7.9	8.1	8.2	6.4	5.8	4.8	4.6		
		MCDBR	4.4	3.9	7.7	14.2	16.8	13.9	12.9	12.4	8.4	5.7	4.1	3.7		
		MCWBR	4.8	3.7	5.5	7.8	9.4	7.3	6.4	6.7	5.7	5.2	4.5	4.0		
		MCWBR	4.4	3.8	7.1	11.9	15.3	12.0	11.6	10.7	6.9	5.5	4.0	3.6		
(35) Mean Daily Temperature Range	5% WB	MCWBR	4.9	3.7	5.4	7.4	9.4	7.1	6.4	6.6	5.6	5.3	4.5	4.0		
	taub		0.295	0.322	0.341	0.375	0.369	0.367	0.395	0.388	0.357	0.342	0.326	0.288		
		taud	2.213	2.271	2.327	2.328	2.421	2.468	2.409	2.415	2.481	2.403	2.237	2.154		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		168	222	254	264	274	276	264	258	248	214	152	128		
		Edh at Noon	15	23	29	34	33	32	33	31	25	20	15	12		
	RadAvg		92	273	758	1325	1837	1961	1837	1442	914	405	120	62		
(45) All-Sky Solar Radiation	RadStd		14	47	78	141	158	100	141	119	106	51	24	9		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (32 neighbors)	+0.63	N/A	N/A	N/A	N/A	N/A	N/A	-207	N/A	N/A

Nomenclature: See separate page